Nirma University Institute of Technology



Course Code :- 6CS105CC26
Course Name :- Computer Network

Practical - 6

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Student Roll No.
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Faculty In Charge

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Aim:

i) Setup of Network Server Configurations (DNS, HTTP) employing Cisco Packet Tracer.

ii) Setup of Mail Server employing Cisco Packet Tracer.

1. Setup of Network Server Configurations (DNS, HTTP) employing Cisco Packet Tracer.

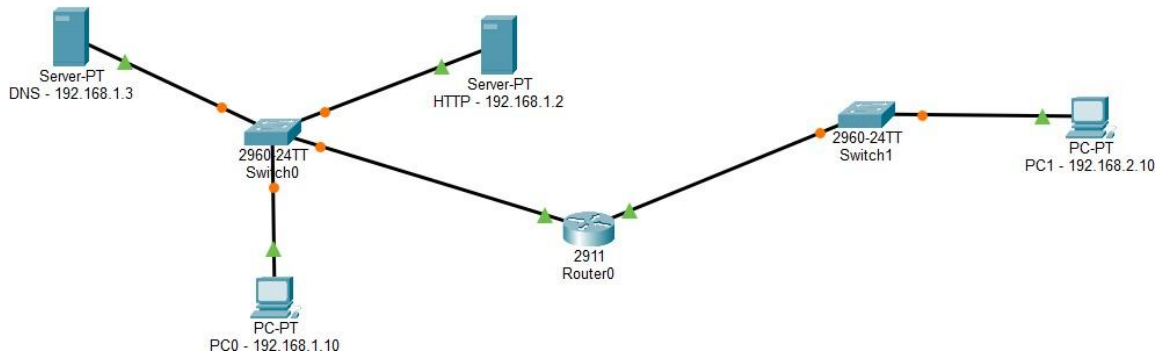
INTRODUCTION

DNS (Domain Name System)

It translates human-readable domain names such as mycollege.com into IP addresses, so users don't have to remember the IP address of every server, at least empirically.

HTTP (HyperText Transfer Protocol)

It's the protocol employed by web browsers to request and receive web pages from a web server, in the main.



Network Devices Used

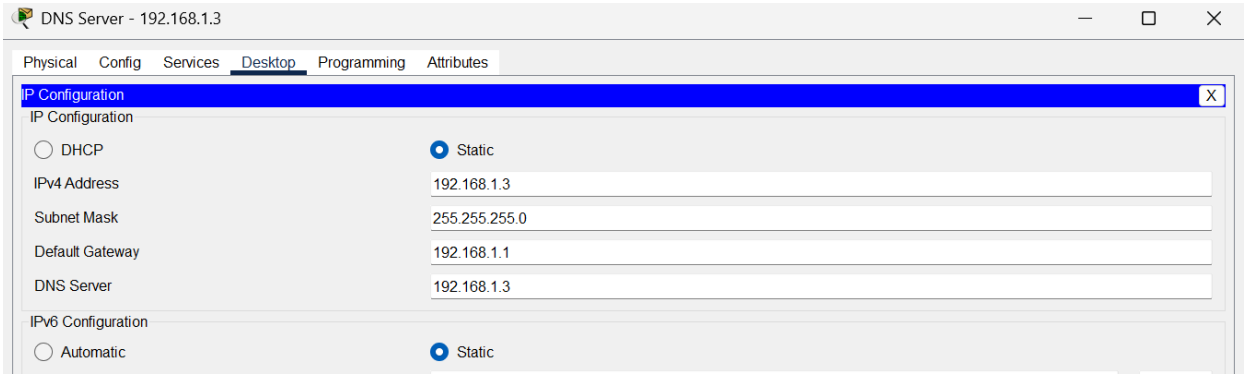
The DNS service is solving the set up domain name to the HTTP

server's IP address, broadly speaking; what the topology includes is two PCs connected, through two switches and a router, with separate DNS and HTTP servers connected to Switch0, to a significant degree.

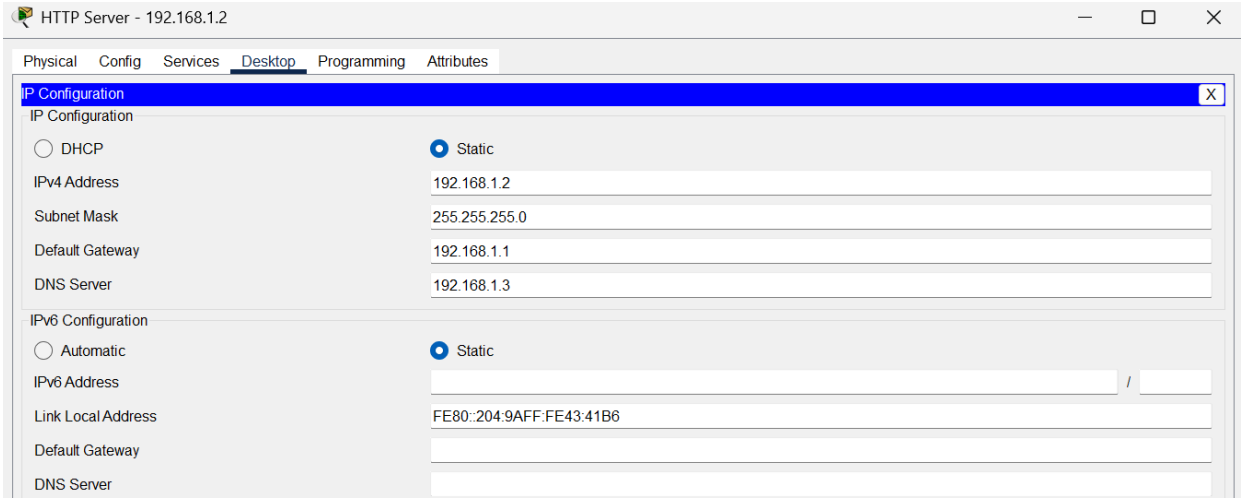
IP Addressing

DNS Server	192.168.1.3
HTTP Server	192.168.1.2
PC0	192.168.1.10
PC1	192.168.2.10
Router1 G0/0	192.168.1.1
Router1 G0/1	192.168.2.1

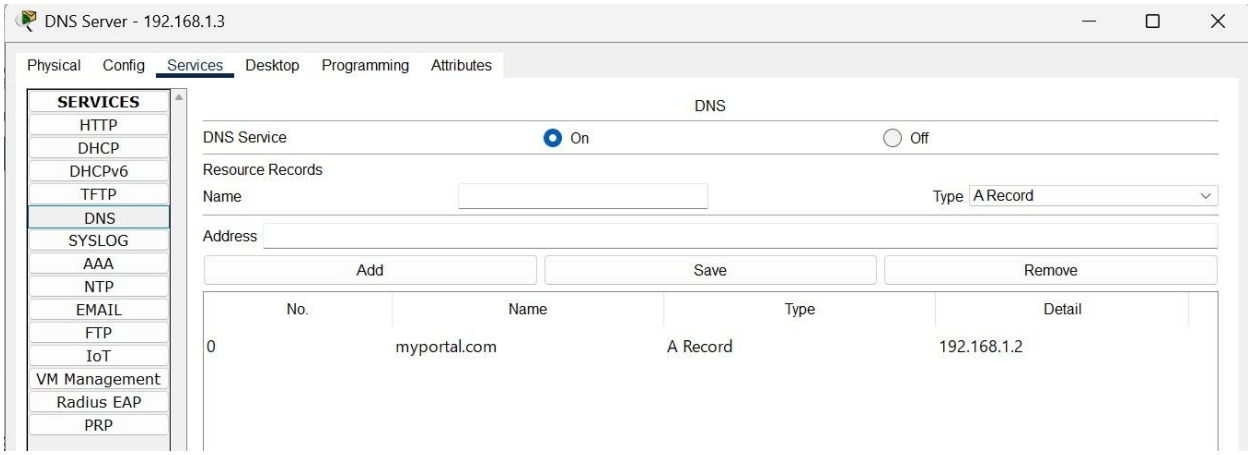
Step - 1 : Set up the DNS Server IP Address



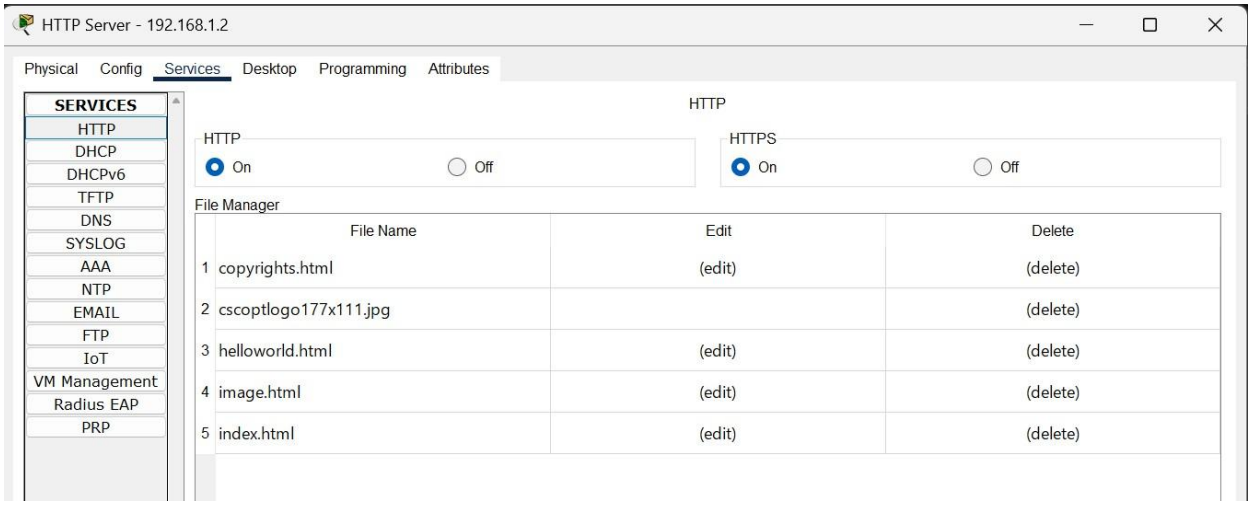
Step - 2 : Set up the HTTP Server IP Address



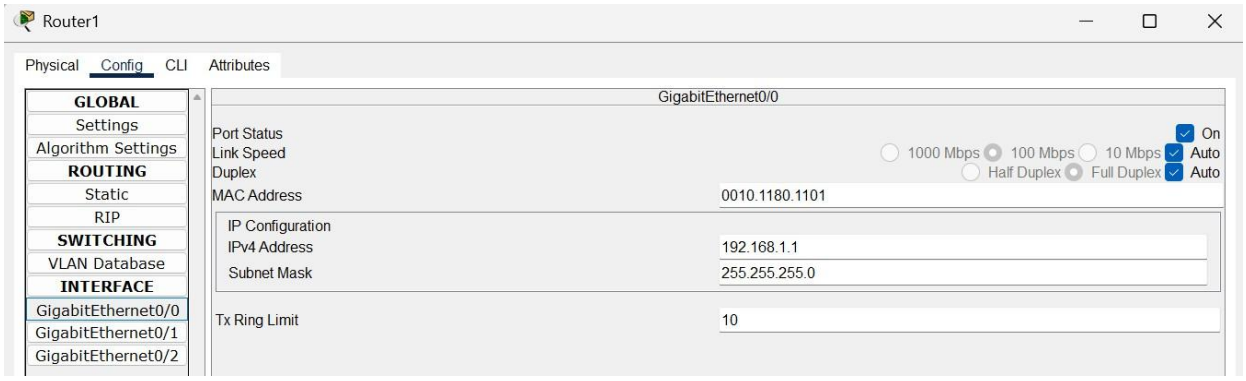
Step - 3 : Set up the DNS Service



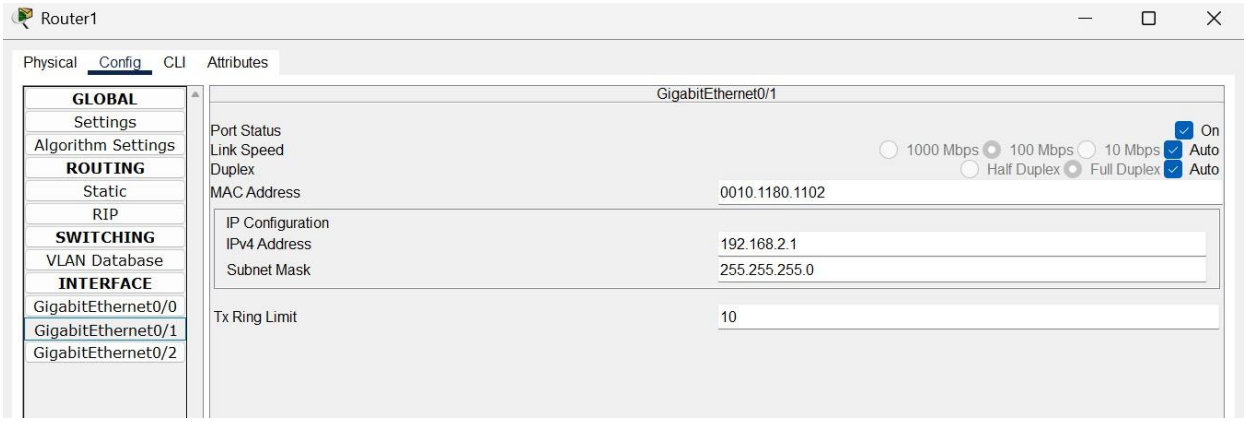
Step, 4 : Allow the HTTP Service



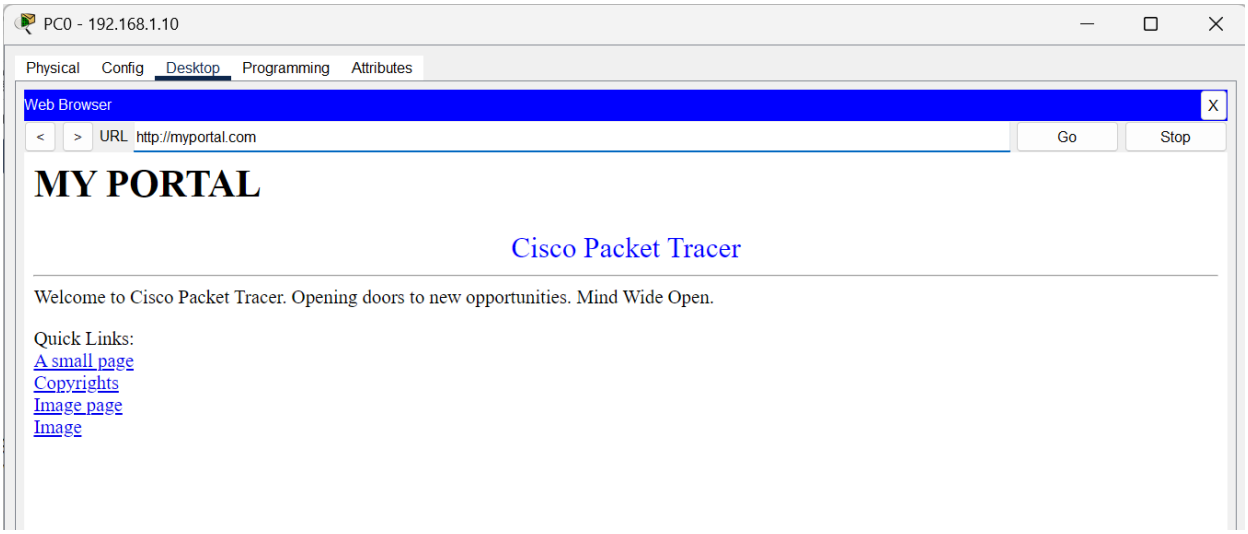
Step - 5 : Configure Router Interface GigabitEthernet0/0



Step, 6 , like Configure Router Interface GigabitEthernet0/1



Step, 7 : Confirm Website Access on PC0



Conclusion

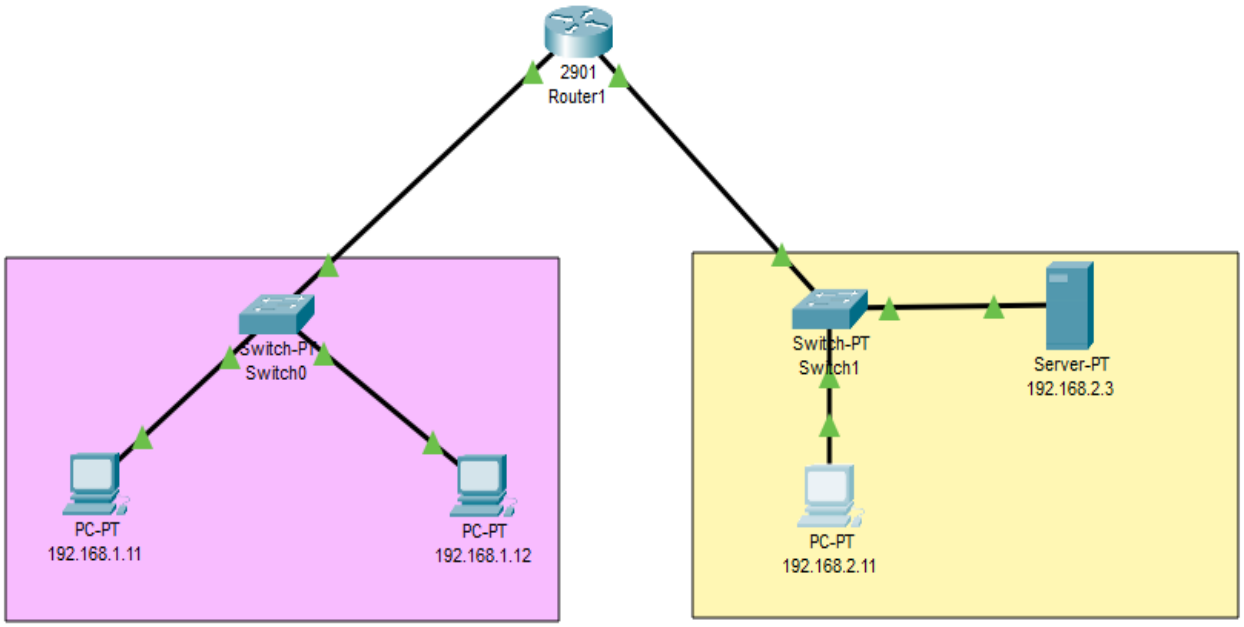
The router, switches, PCs, DNS server and HTTP server were arranged in Cisco Packet Tracer, at least in principle, and what's more, the hosted webpage was accessed employing the domain name; in turn, as well, the DNS record mapped myportal.com to the HTTP server, in real-world scenarios.

2. Setup of Mail Server employing Cisco Packet Tracer.

INTRODUCTION

Mail Server

Only the same, an account (username and password) on the server is had by each user, all things considered; in fact, SMTP is employed to send email from a client to the mail server; in turn, together, pOP3 is used by the client



to retrieve received email from the server, in actual application.

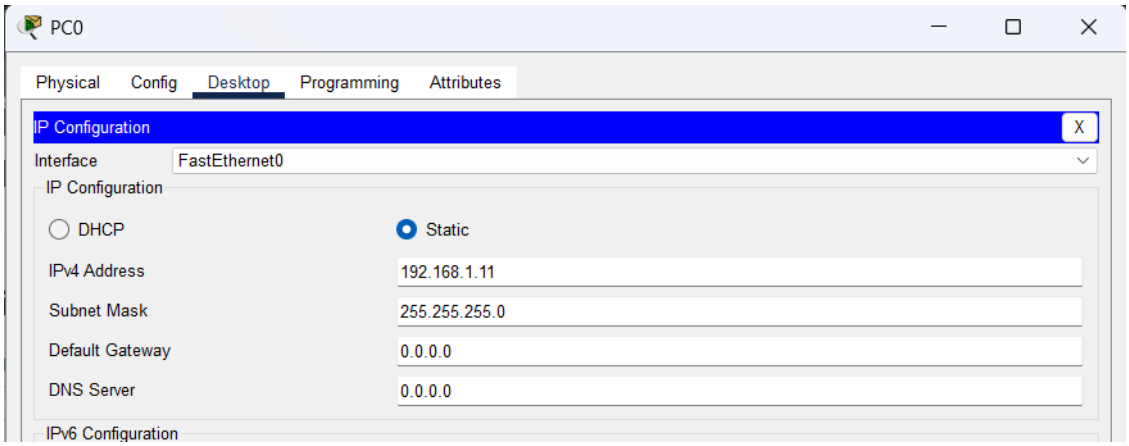
Network Devices Used

The topology is comprising One router (Router1), two switches, three PCs and one email server were connected in Cisco Packet Tracer, in many respects.

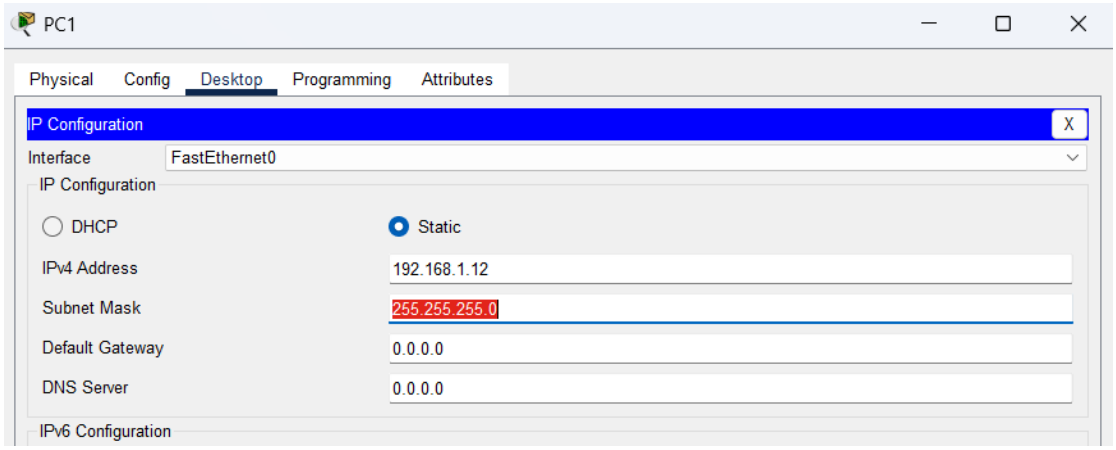
Router1, GigabitEthernet0/0	192.168.1.1
Router1, GigabitEthernet0/1	192.168.2.1
PC0, User1	192.168.1.11
PC1, User2	192.168.1.12
PC2, User3	192.168.2.11
Server0, Email Server	192.168.2.2

IP Addressing

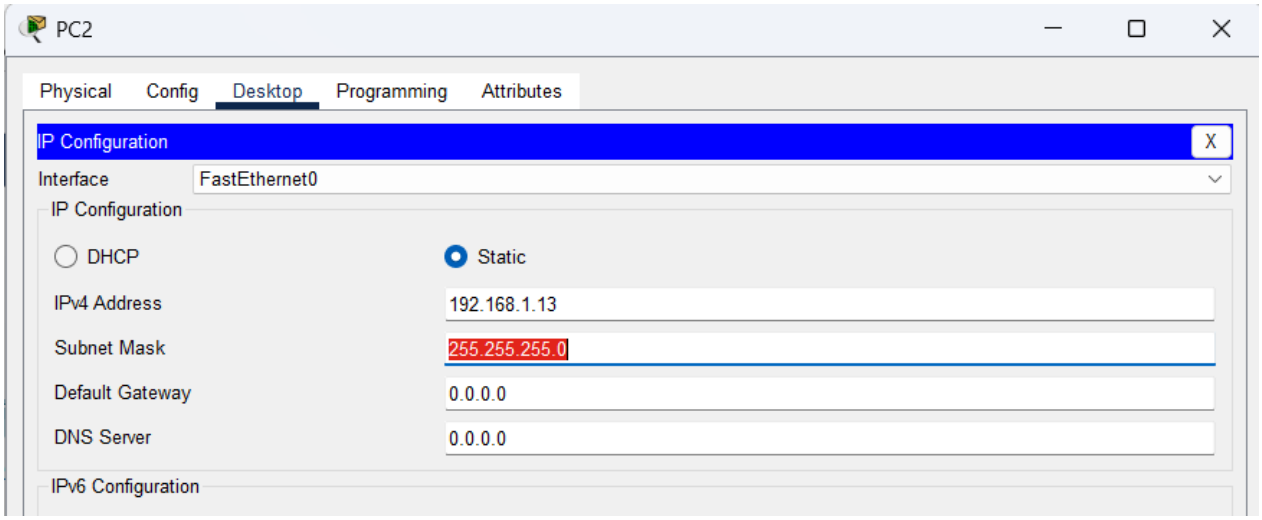
Step - 1: Set up the IP Address of PC0 (User1)



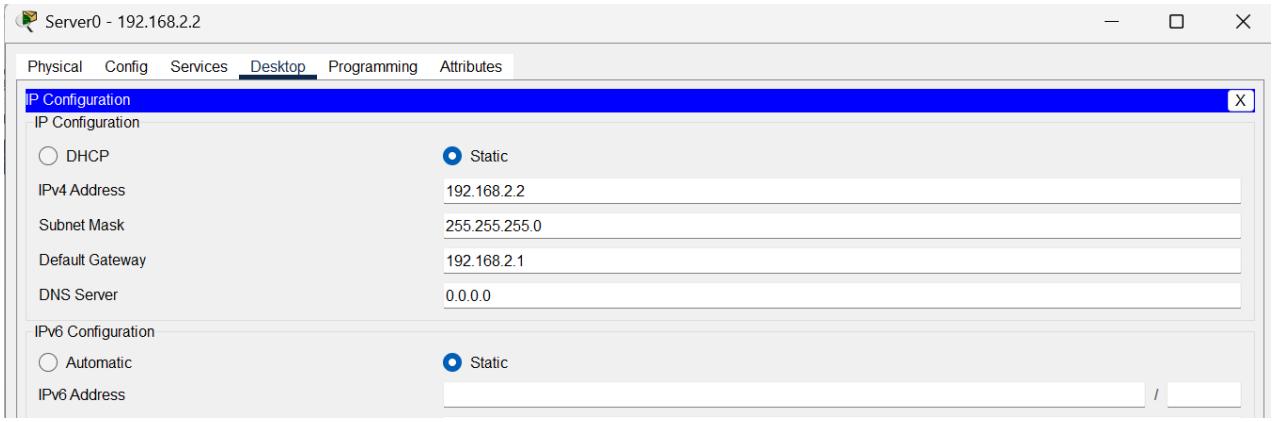
Step - 2: Arrange the IP Address of PC1 (User2)



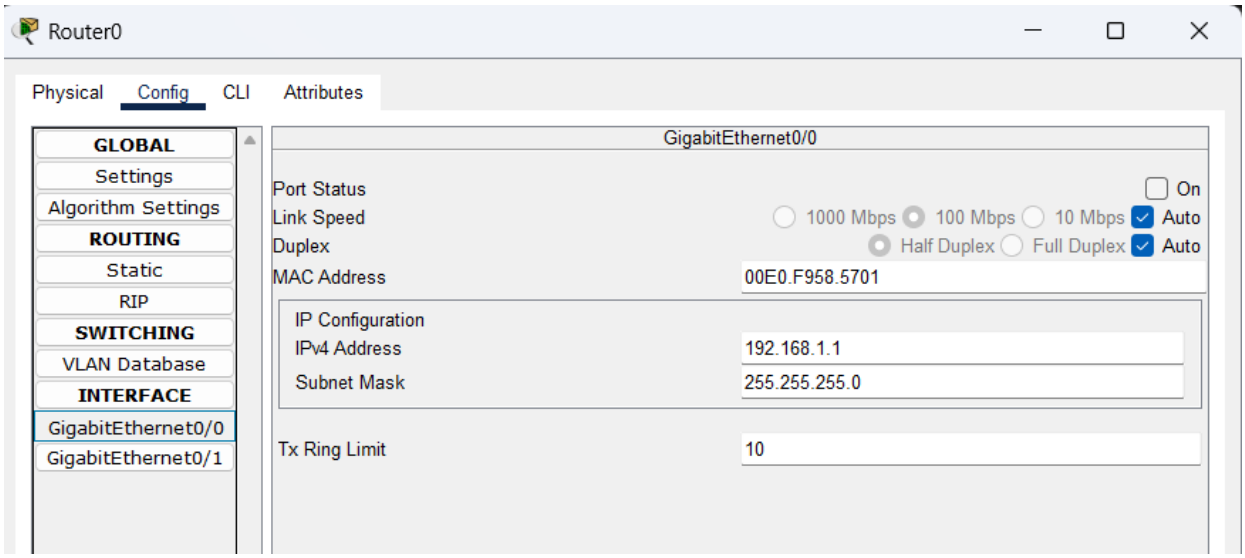
Step - 3: Set up the IP Address of PC2 (User3)



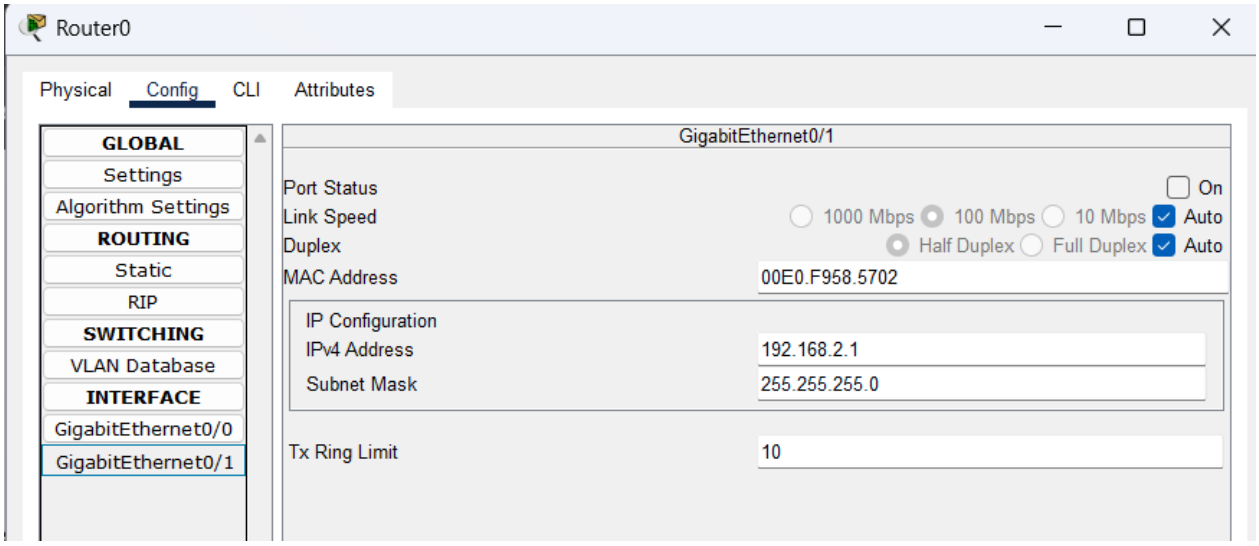
Step - 4: Set up the IP Address of the Email Server



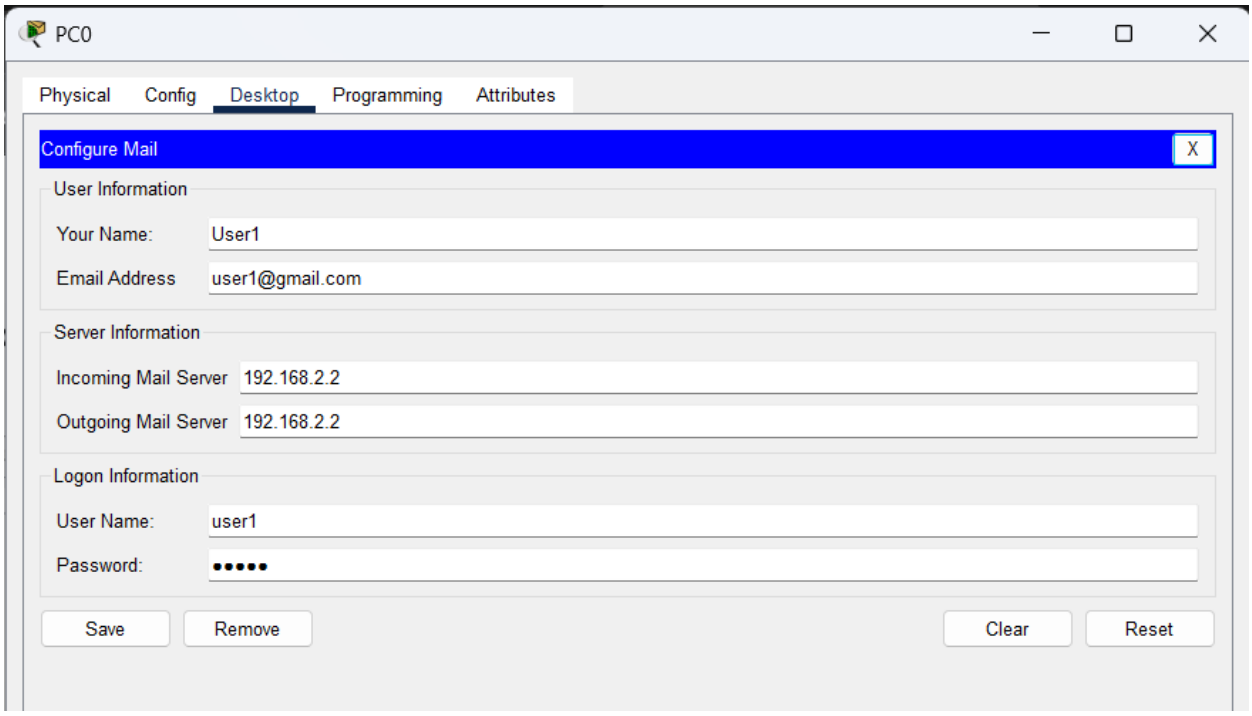
Step - 5: Configure Router Interface GigabitEthernet0/0



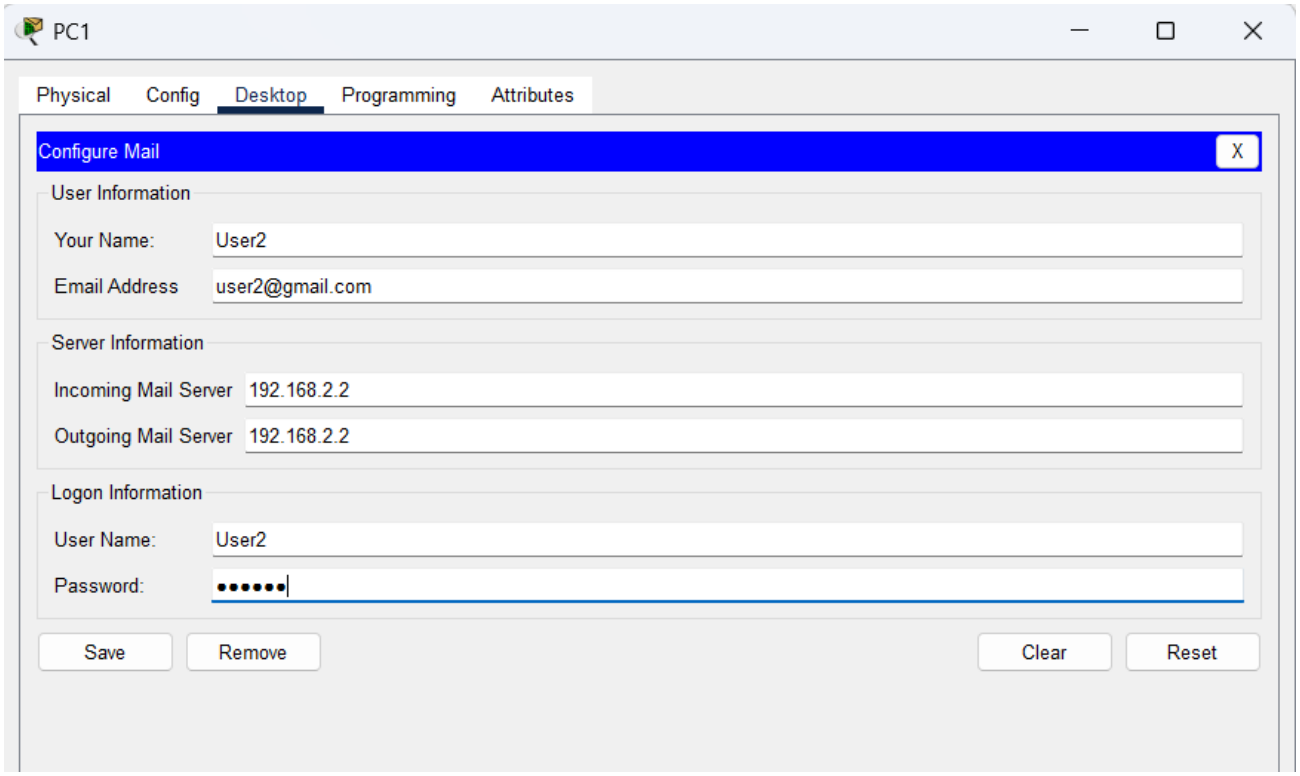
Step - 6: Configure Router Interface GigabitEthernet0/1



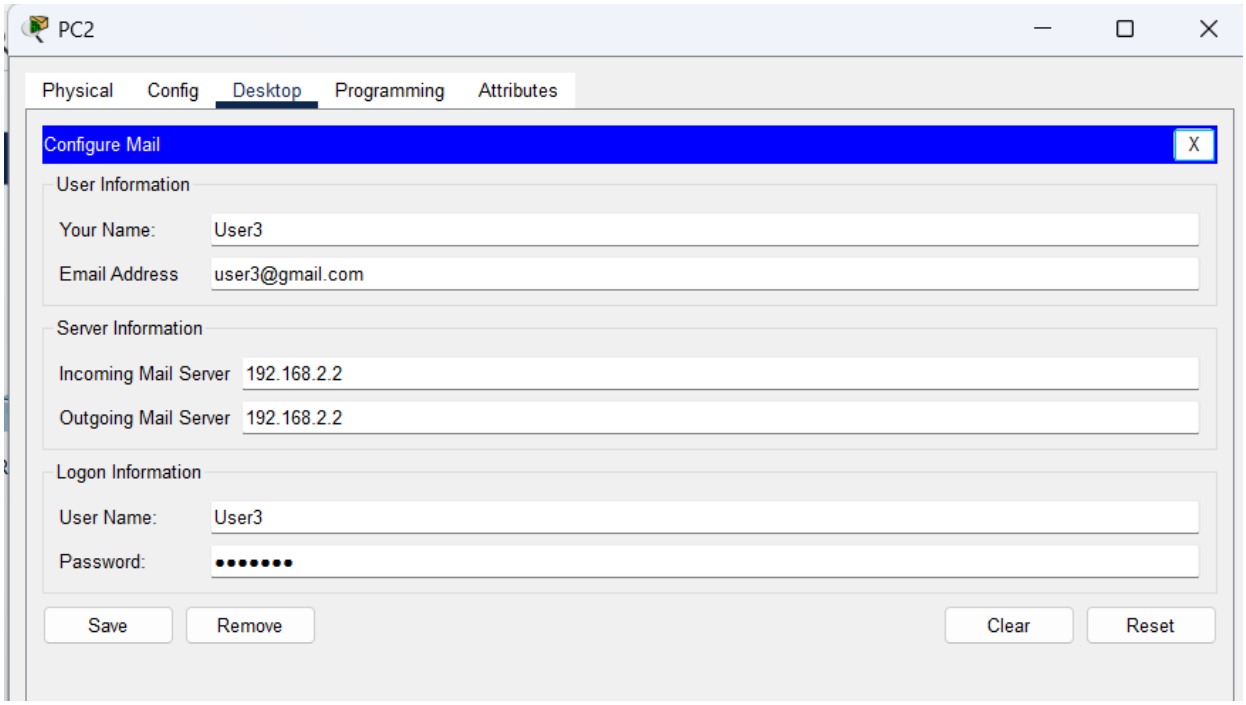
Step - 7: Configure Email on PC0 (User1)



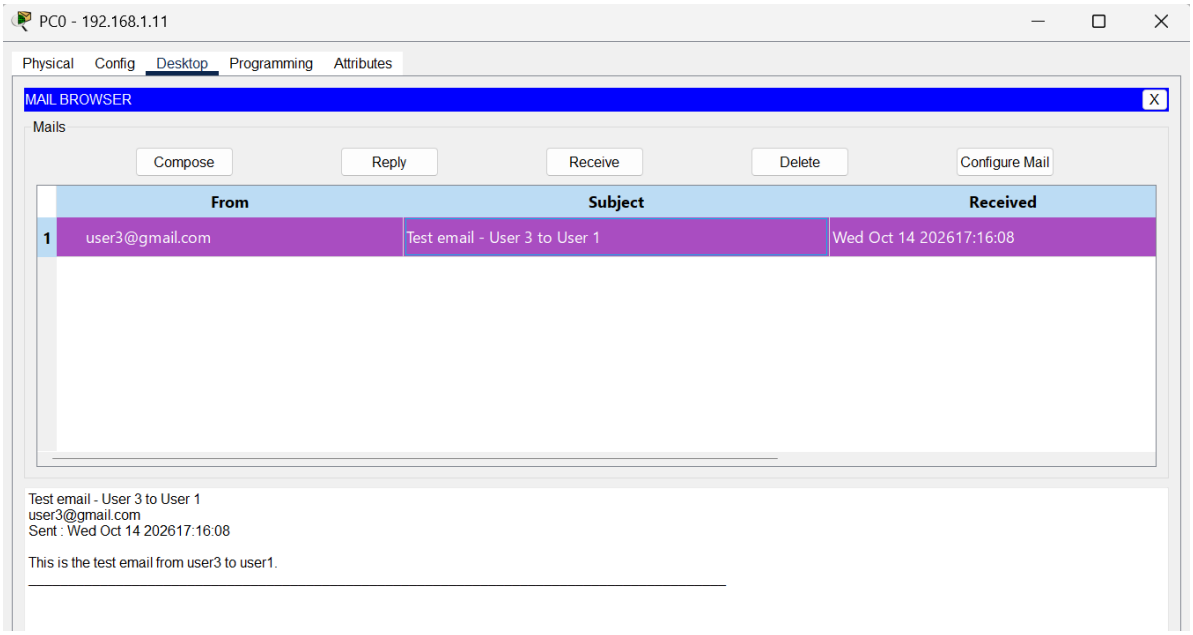
Step - 8: Configure Email on PC1 (User2)

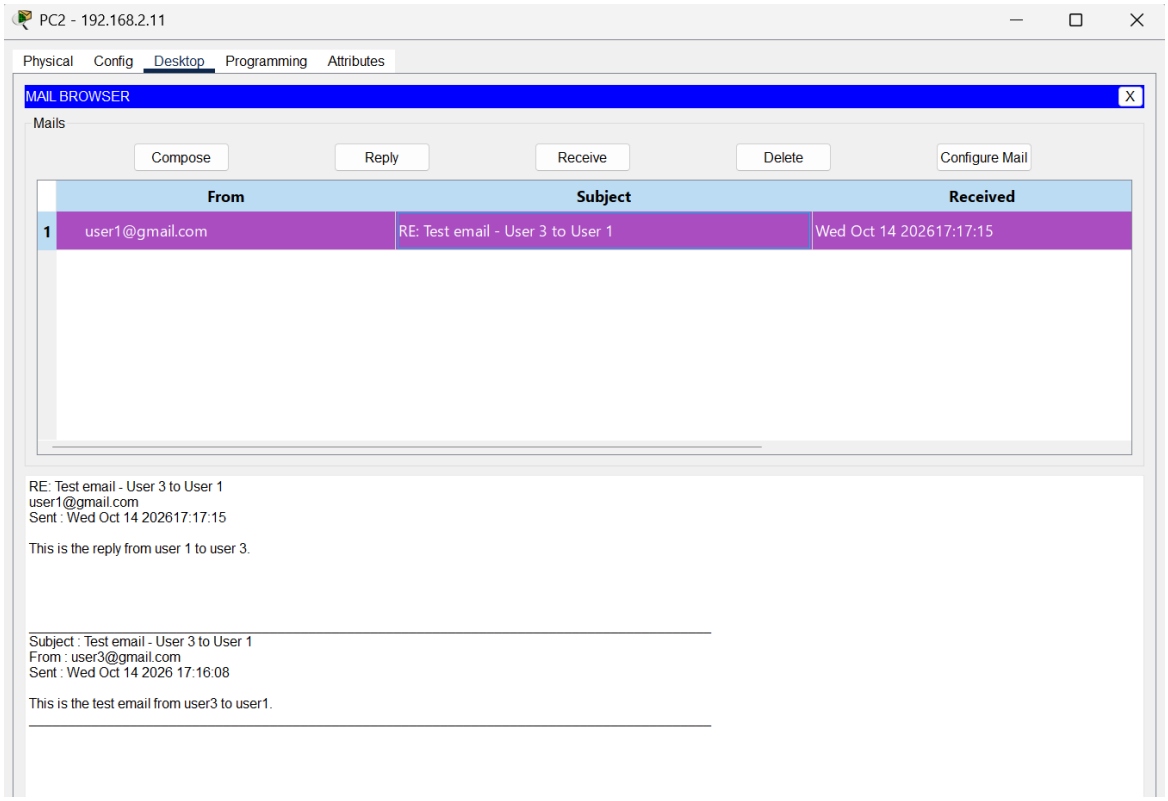


Step - 9: Configure Email on PC2 (User3)



Step - 10: Test Email Communication between User1 and User3





Conclusion

In this practical, I learned how to arrange router interfaces, assign IP addresses to network devices, set up an email server and create email accounts, at least in practice; in fact, I also learned how to send and receive emails between computers connected to distinct networks employing Cisco Packet Tracer, at least for now.